

ASSIGNMENT NO. 4

NAT Gateway Implementation

Objective:

To configure a NAT Gateway in AWS, enable internet access for private subnet instances, and implement secure outbound internet communication using route tables and NAT architecture.

Step 1: Create NAT Gateway

- A NAT Gateway named assignment-4-nat-gateway was created inside the public subnet.
- An Elastic IP address was associated with the NAT Gateway to provide internet connectivity for resources inside the private subnet.

Explanation:

NAT Gateway allows private subnet instances to access the internet securely without assigning public IP addresses directly to the private instances.

The screenshot displays the AWS Management Console interface for a NAT Gateway. At the top, a table lists the NAT gateway 'assignment-4-nat-ga...' with ID 'nat-1f69295ee78113d67', connectivity type 'Public', state 'Available', and route table ID 'rtb-01a2bdb6b...'. Below this, the 'Details' tab is selected for the specific gateway. The details are organized into four columns: NAT gateway ID (nat-1f69295ee78113d67), NAT gateway ARN (arn:aws:ec2:ap-south-1:339333478686:natgateway/nat-1f69295ee78113d67), VPC (vpc-0f9de60bdd13faf04 / assignment-3-vpc), Availability mode (Regional), Connectivity type (Public), Method of EIP allocation (Automatic), State (Available), Created (Tuesday, May 19, 2026 at 18:19:39 GMT+5:30), and State message (-).

Name	NAT gateway ID	Connectivity...	State	State message	Availability ...	Route table ID	P
assignment-4-nat-ga...	nat-1f69295ee78113d67	Public	Available	-	Regional	rtb-01a2bdb6b...	1

nat-1f69295ee78113d67 / assignment-4-nat-gateway				
Details	IP addresses	Monitoring	Flow logs	Tags
Details NAT gateway ID nat-1f69295ee78113d67 NAT gateway ARN arn:aws:ec2:ap-south-1:339333478686:natgateway/nat-1f69295ee78113d67 VPC vpc-0f9de60bdd13faf04 / assignment-3-vpc				

Step 2: Update Private Route Table

The private route table was updated by adding a route:

0.0.0.0/0 → NAT Gateway

This route redirects internet-bound traffic from the private subnet to the NAT Gateway.

Explanation:

The route table controls traffic flow inside the VPC. By pointing internet traffic to the NAT Gateway, private EC2 instances can access the internet without becoming publicly accessible.

The screenshot displays the AWS Management Console interface for Route Tables. The top section shows a list of route tables, with the 'private-route-table' selected. Below this, the 'Routes' tab is active, showing two routes: a default route (0.0.0.0/0) pointing to a NAT Gateway and a local route (10.0.0.0/16) pointing to the local network.

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
-	rtb-0cc74e8a63ed00360	-	-	Yes	vpc-01eaa90696ea8e9a9
-	rtb-0f5e20220d2c99d66	-	-	Yes	vpc-0f9de60bdd13faf04 assi
☒ private-route-table	rtb-0fbc7510c12c18345	2 subnets	-	No	vpc-0f9de60bdd13faf04 assi
☐ RDS-Pvt-rt	rtb-089f6ba0d3d63bfb8	3 subnets	-	No	vpc-01eaa90696ea8e9a9

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	nat-1f69295ee78113d67	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

Step 3: Launch Public EC2 Instance

- A public EC2 instance was launched inside the public subnet.
- This instance acted as a bastion host to securely connect to the private EC2 instance.

Explanation:

A bastion host is a public server used to access private servers securely inside a private subnet.

The public EC2 instance was launched inside the public subnet with internet access through the Internet Gateway.

The screenshot displays the AWS Management Console interface for the 'Instances' page. At the top, there's a header with 'Instances (1/4)' and a search bar. Below this, a table lists the instances. The first instance, 'public-ec2' with ID 'i-088108adcdf3179a3', is in a 'Running' state. The 'Networking' tab is selected for this instance, showing details like VPC ID, Subnet ID, Availability zone, and IP addresses.

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
public-ec2	i-088108adcdf3179a3	Running	t3.micro	3/3 checks passed	ap-south-1a	-

i-088108adcdf3179a3 (public-ec2)		
Details	Status and alarms	Monitoring
VPC ID vpc-0f9de60bdd13faf04 (assignment-3-vpc)	Subnet ID subnet-0eb710d22440747a7 (public-subnet-az1)	Availability zone ap-south-1a
Availability zone ID aps1-az1	Outpost ID -	
IP addresses		
Public IPv4 address 65.0.103.236 open address	Private IPv4 addresses 10.0.1.252	IPv6 addresses -
Secondary private IPv4 addresses -	Carrier IP addresses (ephemeral) -	

Step 4: Launch Private EC2 Instance

- A private EC2 instance was launched inside the private subnet without assigning a public IP address.
- The instance used the NAT Gateway for outbound internet access.

Explanation:

Private EC2 instances are not directly accessible from the internet. They use NAT Gateway for secure outbound internet communication.

The screenshot displays the AWS Management Console interface for EC2 instances. At the top, there's a header for 'Instances (1/4)' with a search bar and various action buttons like 'Connect', 'Instance state', 'Actions', and 'Launch instances'. Below this, a table lists two instances: 'private-ec2' and 'public-ec2'. The 'private-ec2' instance is highlighted, showing its ID 'i-0d519d9113bb3e58e', state 'Running', type 't3.micro', and availability zone 'ap-south-1a'. Below the table, the 'Networking' tab for the 'private-ec2' instance is expanded, showing details such as VPC ID, Subnet ID, Availability zone, and IP addresses. The instance has a private IPv4 address of 10.0.11.5 and no public IPv4 address.

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
private-ec2	i-0d519d9113bb3e58e	Running	t3.micro	3/3 checks passed	ap-south-1a	-
public-ec2	i-088108adcdf3179a3	Running	t3.micro	3/3 checks passed	ap-south-1a	-

i-0d519d9113bb3e58e (private-ec2)

Networking

VPC ID vpc-0f9de60bdd13faf04 (assignment-3-vpc)	Subnet ID subnet-0ada64a7923c704de	Availability zone ap-south-1a
Availability zone ID aps1-az1	Outpost ID -	
Public IPv4 address -	Private IPv4 addresses 10.0.11.5	IPv6 addresses -
Secondary private IPv4 addresses -	Carrier IP addresses (ephemeral) It is taking a bit longer than usual to fetch your data	

Step 5: Connect to Private EC2 Using SSH

The public EC2 instance was used as a bastion host to SSH into the private EC2 instance using the private IP address.

Explanation:

Since private EC2 instances do not have public IP addresses, they cannot be accessed directly from the internet. SSH access is performed through the public EC2 instance.

```
ubuntu@ip-10-0-1-252:~$ ssh -i assignment1-key.pem ubuntu@10.0.11.5
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1004-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue May 19 14:51:05 UTC 2026

System load:  0.0           Temperature:   -273.1 C
Usage of /:   30.4% of 6.61GB Processes:    117
Memory usage: 25%          Users logged in: 0
Swap usage:   0%           IPv4 address for ens5: 10.0.11.5

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ubuntu@ip-10-0-11-5:~$ curl https://api.ipify.org
```

Step 6: Test Internet Connectivity

Internet connectivity from the private EC2 instance was tested using the command:

```
curl https://api.ipify.org
```

The command returned a public IP address, confirming successful internet connectivity through the NAT Gateway.

Explanation:

The private EC2 instance accessed the internet securely without having a public IP address.

```
ubuntu@ip-10-0-11-5:~$ curl https://api.ipify.org
15.207.200.99ubuntu@ip-10-0-11-5:~$ sudo apt update
```

Step 7: Test Package Updates

The command:

```
sudo apt update
```

was executed successfully inside the private EC2 instance, confirming that outbound internet access through the NAT Gateway was functioning properly.

```
15.207.200.99ubuntu@ip-10-0-11-5:~$ sudo apt update
Get:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease [136 kB]
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease [136 kB]
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease [136 kB]
Get:4 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 Packages [1483 kB]
Get:5 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/main Translation-en [524 kB]
Get:6 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64 Components [395 kB]
Get:7 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64 c-n-f Metadata [32.4 kB]
Get:8 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 Packages [16.3 MB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security InRelease [136 kB]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/main amd64v3 Packages [42.3 kB]
Get:11 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [19.1 kB]
Get:12 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [2844 B]
Get:13 http://security.ubuntu.com/ubuntu resolute-security/main amd64 c-n-f Metadata [968 B]
Get:14 http://security.ubuntu.com/ubuntu resolute-security/universe amd64v3 Packages [40.7 kB]
Get:15 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [13.3 kB]
Get:16 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/universe Translation-en [6329 kB]
Get:17 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [20.6 kB]
Get:18 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 c-n-f Metadata [628 B]
Get:19 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64v3 Packages [23.7 kB]
Get:20 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [5228 B]
Get:21 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 Components [212 B]
Get:22 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 c-n-f Metadata [120 B]
Get:23 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64 Components [4556 kB]
Get:24 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64 c-n-f Metadata [313 kB]
Get:25 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/restricted amd64v3 Packages [152 kB]
Get:26 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/restricted amd64 Components [556 B]
Get:27 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/multiverse amd64v3 Packages [291 kB]
Get:28 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/multiverse Translation-en [127 kB]
Get:29 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/multiverse amd64 Components [50.0 kB]
Get:30 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/multiverse amd64 c-n-f Metadata [8276 B]
Get:31 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 Packages [78.1 kB]
Get:32 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [24.2 kB]
Get:33 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [2844 B]
Get:34 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64 c-n-f Metadata [1132 B]
Get:35 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64v3 Packages [40.7 kB]
Get:36 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [13.3 kB]
Get:37 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [20.6 kB]
Get:38 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 c-n-f Metadata [624 B]
Get:39 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/restricted amd64v3 Packages [174 kB]
Get:40 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [29.3 kB]
Get:41 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Components [216 B]
Get:42 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 c-n-f Metadata [116 B]
Get:43 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports/main amd64 Components [212 B]
Get:44 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports/main amd64 c-n-f Metadata [112 B]
Get:45 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [216 B]
Get:46 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports/universe amd64 c-n-f Metadata [116 B]
Get:47 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports/restricted amd64 Components [216 B]
Get:48 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports/restricted amd64 c-n-f Metadata [120 B]
Get:49 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports/multiverse amd64 Components [216 B]
Get:50 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports/multiverse amd64 c-n-f Metadata [120 B]
Fetched 31.6 MB in 4s (7940 kB/s)
28 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

Working Flow of NAT Gateway:

1. Private EC2 sends internet request
2. Route table redirects traffic to NAT Gateway
3. NAT Gateway forwards traffic to Internet Gateway
4. Internet response returns through NAT Gateway
5. NAT Gateway sends response back to private EC2

This allows private instances to access the internet securely without exposing them directly to public networks.

Key Learnings –

- Learned NAT Gateway implementation
- Understood private subnet internet access
- Learned route table configuration
- Understood bastion host architecture
- Learned secure VPC networking concepts
- Understood outbound internet access for private instances

Conclusion:

Successfully implemented NAT Gateway architecture in AWS. Private EC2 instances were able to securely access the internet through the NAT Gateway without using public IP addresses.